

Spreading Markushevich Bases in Banach Spaces are Schauder

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Status. Full answer, likely valid. The source paper asks whether there exists a spreading Markushevich basis in a Banach space which fails to be a Schauder basis. The answer is no.

Source Question

The source is J. L. Ansorena and A. Marcos, *Remarks about symmetry-type conditions of conditional bases of Banach spaces*, arXiv:2601.15784. On page 16, after discussing the failure of the symmetric Banach-space theorem in the locally non-convex quasi-Banach setting and the summing basis of c_0 , the authors write:

“In contrast, it is unclear to the authors whether a spreading Markushevich basis that fails to be a Schauder basis exists.”

for all $A, B \in \mathcal{P}_{<\infty}(\mathbb{N})$, and all $(\varepsilon_n)_{n \in A}, (\delta_n)_{n \in B}$ in $S_{\mathbb{F}}$.

We point out that Theorem 3.6 breaks down in the locally non-convex setting. Indeed, the authors of [3] constructed a quasi-Banach space $\mathbb{X}$ with a conditional symmetric Markushevich basis. This space $\mathbb{X}$ is locally p -convex for all $0 < p < 1$, and its canonical basis is almost greedy, then unconditional for constant coefficients. We remark that this example from [3] does not rule out possible relations between totality and weakened forms of unconditionality, such as quasi-greediness or unconditionality for constant coefficients, in the general framework of quasi-Banach spaces. The summing basis of c_0 witnesses that the spreading counterpart of Theorem 3.6 does not hold. In contrast, it is unclear to the authors whether a spreading Markushevich basis that fails to be a Schauder basis exists.

STATEMENTS AND DECLARATIONS

Conflict of interest. The authors have no competing interests to declare that are relevant to the content of this article.

Definitions and Source Facts

We use the terminology of the source paper. A complete minimal system $(x_n)_{n=1}^{\infty}$ in a Banach space X , with coordinate functionals $(x_n^*)_{n=1}^{\infty}$, is a Markushevich basis when it is total, equivalently when

$$x_n^*(x) = 0 \quad \text{for all } n \in \mathbb{N} \quad \implies \quad x = 0.$$

It is a spreading basis when every increasing map $\psi : \mathbb{N} \rightarrow \mathbb{N}$ induces a bounded shifting operator L_ψ , $L_\psi x_n = x_{\psi(n)}$, which is an isomorphism onto its range.

We will use two facts from the source paper. First, every spreading basis in a quasi-Banach space, hence in a Banach space, is seminormalized and M-bounded [1, Corollary 2.10]. Second, for every increasing ψ , the operator L_ψ identifies the original system with the subsequence $(x_{\psi(n)})$. Thus the original sequence is equivalent, as a candidate Schauder basis, to each of its subsequences.

Proof Intuition

The only possible way for a bounded non-compact Banach-space sequence to avoid having an immediate basic subsequence is through weak compactness phenomena. For a spreading Markushevich basis those phenomena are harmless. Rosenthal's ℓ_1 -theorem gives either an ℓ_1 -subsequence or a weak-Cauchy subsequence. An ℓ_1 -subsequence is already basic. A non-trivial weak-Cauchy subsequence has a basic subsequence by the standard Bessaga-Pelczynski selection argument. If the weak-Cauchy subsequence actually converges weakly in X , totality forces its weak limit to be zero, because every coordinate functional eventually vanishes on that subsequence. It is then weakly null and seminormalized, so the Bessaga-Pelczynski selection principle again gives a basic subsequence. Once any basic subsequence exists, spreading transfers the uniformly bounded partial-sum projections back to the original complete minimal system.

Theorem 1. *Let X be a Banach space and let $(x_n)_{n=1}^\infty$ be a spreading Markushevich basis of X in the sense of Ansorena and Marcos. Then $(x_n)_{n=1}^\infty$ is a Schauder basis of X . Consequently there is no spreading Markushevich basis in a Banach space which fails to be a Schauder basis.*

Proof. By the source corollary cited above, $(x_n)_{n=1}^\infty$ is bounded and bounded away from zero. We first show that $(x_n)_{n=1}^\infty$ has a Schauder basic subsequence.

Apply Rosenthal's ℓ_1 -theorem to $(x_n)_{n=1}^\infty$. If some subsequence is equivalent to the unit vector basis of ℓ_1 , then it is a Schauder basic sequence and there is nothing more to prove at this stage. Otherwise, $(x_n)_{n=1}^\infty$ has a weak-Cauchy subsequence, say $y_j = x_{k_j}$, with $k_1 < k_2 < \dots$.

If (y_j) is non-trivial weak-Cauchy, meaning that its weak-star limit in X^{**} is not an element of X , then it has a Schauder basic subsequence. This is the standard non-trivial weak-Cauchy selection theorem; for instance, it is the $l = 1$ case of [2, Proposition 2.4], where the condition $X \cap \text{span}\{y^{**}\} = \{0\}$ holds automatically because $y^{**} \notin X$.

It remains to consider the case in which (y_j) converges weakly to some $y \in X$. Fix $n \in \mathbb{N}$. Since the integers k_j are strictly increasing, $k_j \neq n$ for all sufficiently large j . Hence

$$x_n^*(y) = \lim_j x_n^*(x_{k_j}) = 0.$$

This holds for every n . Since $(x_n)_{n=1}^\infty$ is Markushevich, its coordinate functionals are total, and therefore $y = 0$. Thus (y_j) is a seminormalized weakly null sequence. By the Bessaga-Pelczynski selection principle, (y_j) has a Schauder basic subsequence.

We have proved that there are indices $n_1 < n_2 < \dots$ such that $z_j = x_{n_j}$ is a Schauder basic sequence for its closed linear span $Z = [z_j]$. Let P_N^Z denote its N -th partial-sum projection, so $\sup_N \|P_N^Z\| < \infty$.

Let $\psi(j) = n_j$. By spreadingness, $T = L_\psi$ is an isomorphism from X onto Z with $Tx_j = z_j$. For finite linear combinations,

$$TS_N x = P_N^Z T x,$$

where S_N is the N -th initial coordinate projection associated with $(x_n)_{n=1}^\infty$. Passing to the closure gives

$$S_N = T^{-1}P_N^Z T, \quad N \in \mathbb{N}.$$

Consequently

$$\sup_N \|S_N\| \leq \|T^{-1}\| \|T\| \sup_N \|P_N^Z\| < \infty.$$

Because $(x_n)_{n=1}^\infty$ is complete, uniformly bounded initial partial-sum projections make $(x_n)_{n=1}^\infty$ a Schauder basis for X . This proves the theorem. $\square$

Verification and Novelty Check

The proof uses only standard Banach-space selection tools plus the source paper’s own spreading-basis facts. The totality hypothesis is used exactly once, to force any weak limit of a subsequence to be zero.

Cheap local indexes were searched for arXiv:2601.15784 and the phrases *spreading Markushevich*, *Markushevich basis*, *spreading basis*, and *summing basis*. No existing packet for this exact question was found. Local source searches found related conditional-spreading-basis literature, especially Freeman–Odell–Sari–Zheng [3], but that paper defines a spreading basis as a basic sequence and therefore does not itself answer the present non-Schauder Markushevich-basis question. Web searches on June 16, 2026 for exact and close phrases including *spreading Markushevich basis*, *spreading M-basis Schauder*, *complete minimal spreading basis Schauder*, and the exact sentence from the source did not find a direct prior answer.

Human Review Recommendation

Review should check the interpretation of the source’s word “isomorphism” for the shifting operator L_ψ : the proof needs it as an isomorphism onto the closed span of the subsequence, which is how the source’s translation and inverse-shift discussion is used. Review should also check the cited non-trivial weak-Cauchy selection step. If both points are accepted, the result is a full negative answer to the source question.

References

- [1] J. L. Ansorena and A. Marcos, *Remarks about symmetry-type conditions of conditional bases of Banach spaces*, arXiv:2601.15784, 2026.
- [2] S. A. Argyros, P. Georgiou, A. Lagos, and P. Motakis, *Joint spreading models and uniform approximation of bounded operators*, arXiv:1712.07638, 2017.
- [3] D. Freeman, E. Odell, B. Sari, and B. Zheng, *On spreading sequences and asymptotic structures*, arXiv:1607.03587, 2016.
- [4] H. P. Rosenthal, *A characterization of Banach spaces containing ℓ_1* , Proc. Nat. Acad. Sci. USA 71 (1974), 2411–2413.
- [5] C. Bessaga and A. Pelczynski, *On bases and unconditional convergence of series in Banach spaces*, Studia Math. 17 (1958), 151–164.